



Mini Project

RT 524- Digital Agriculture and Advanced Analytics

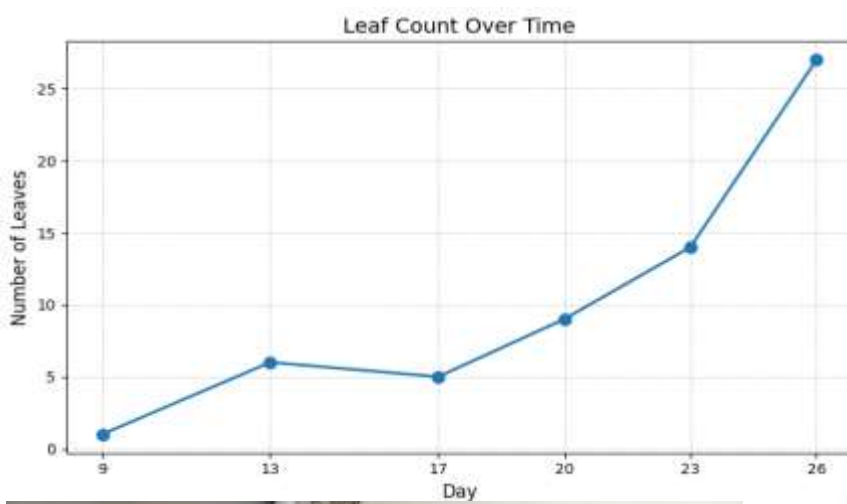
Supervisor-

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SART, IITG

12-07-2026

Presented by-

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SART. IITG



AUTOMATED PLANT TRAIT EXTRACTION FROM IMAGES (PLANT PHENOMICS)

Plant phenomics: Study of measuring and analyzing plant characteristics (traits) using digital tools and technology

OBJECTIVES

Develop an **automated plant analysis pipeline**

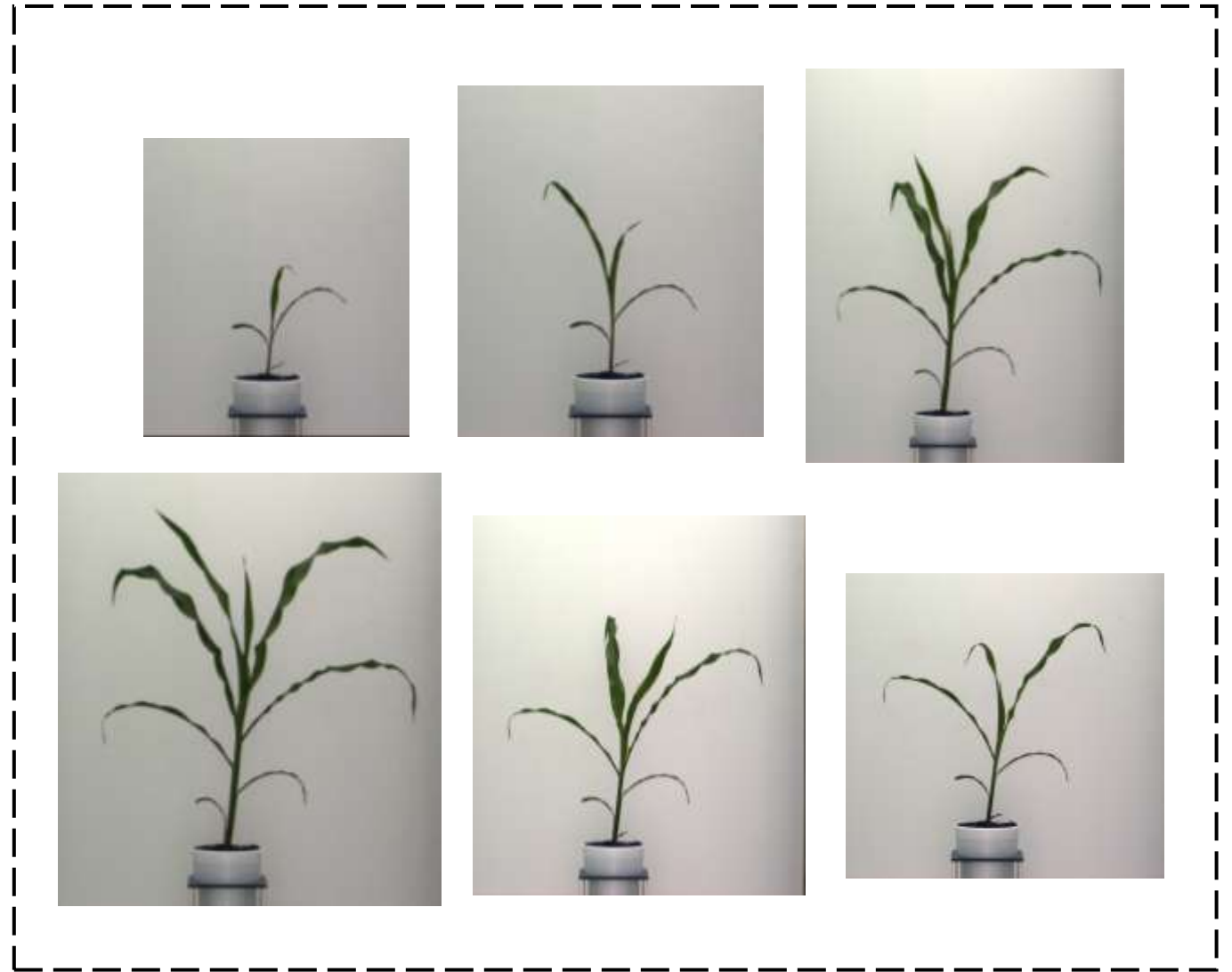
Measure:

- Leaf count
- Leaf length
- Growth trends



RECOURSES

Google
colab

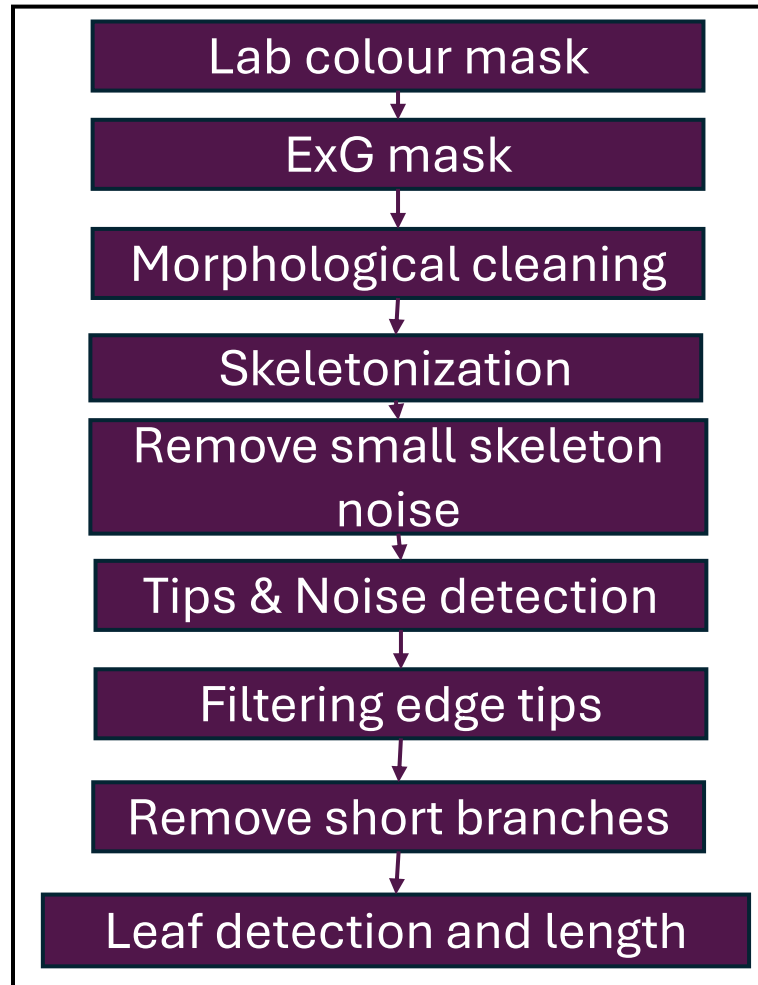


Ref & Credit: Plant Vision Initiative/UNL

METHODOLOGY

Batch processing

Image Processing



**Bounding
boxes**



MAJOR PROCESSES

IMAGE SEGMENTATION

Separating plant from background

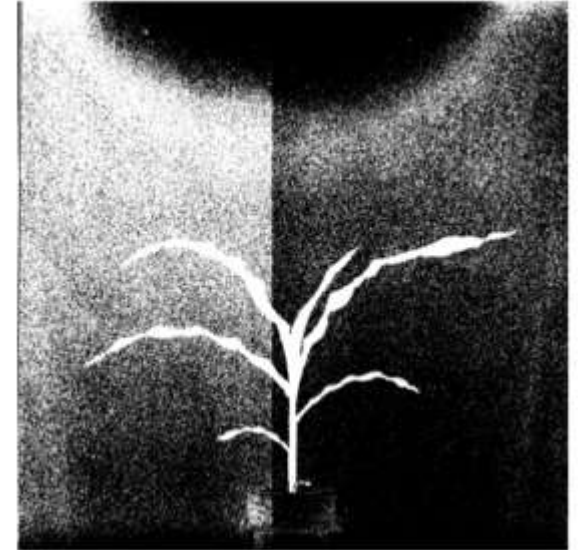
Methods Used:

- **Lab Color Space:** Uses green channel properties
- **Excess Green Index (ExG):** Highlights vegetation

Day_023 — Original



Lab Mask



Day_023
ExG Mask



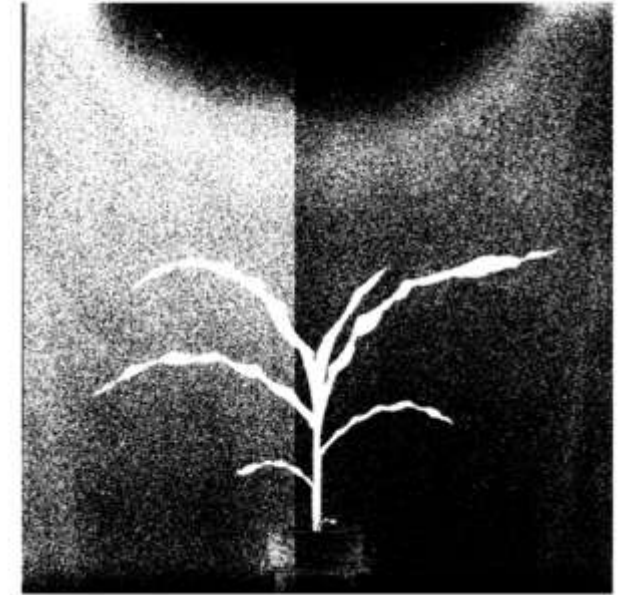
SKELETONIZATION

- Reduce plant to a **1-pixel-wide structure**
- Extract plant structure (centerline)
- Enables:
 - Leaf length calculation
 - Tip detection

Day_023 — Original



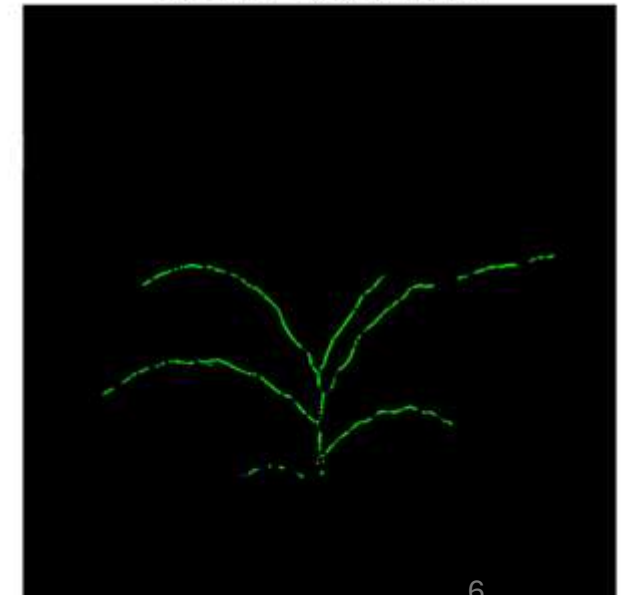
Lab Mask

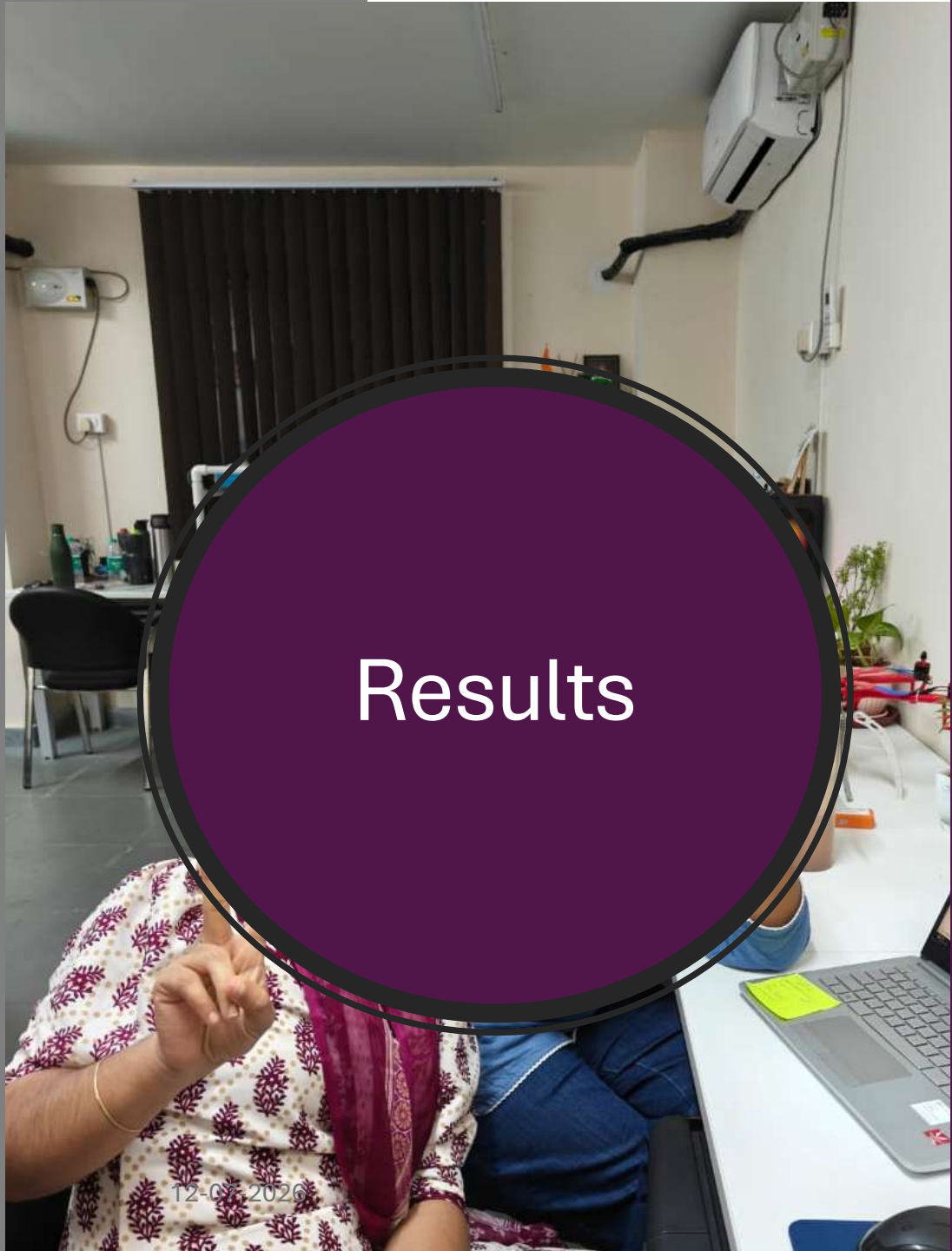


Day_023
ExG Mask



Skeleton + Tips (15 leaves)

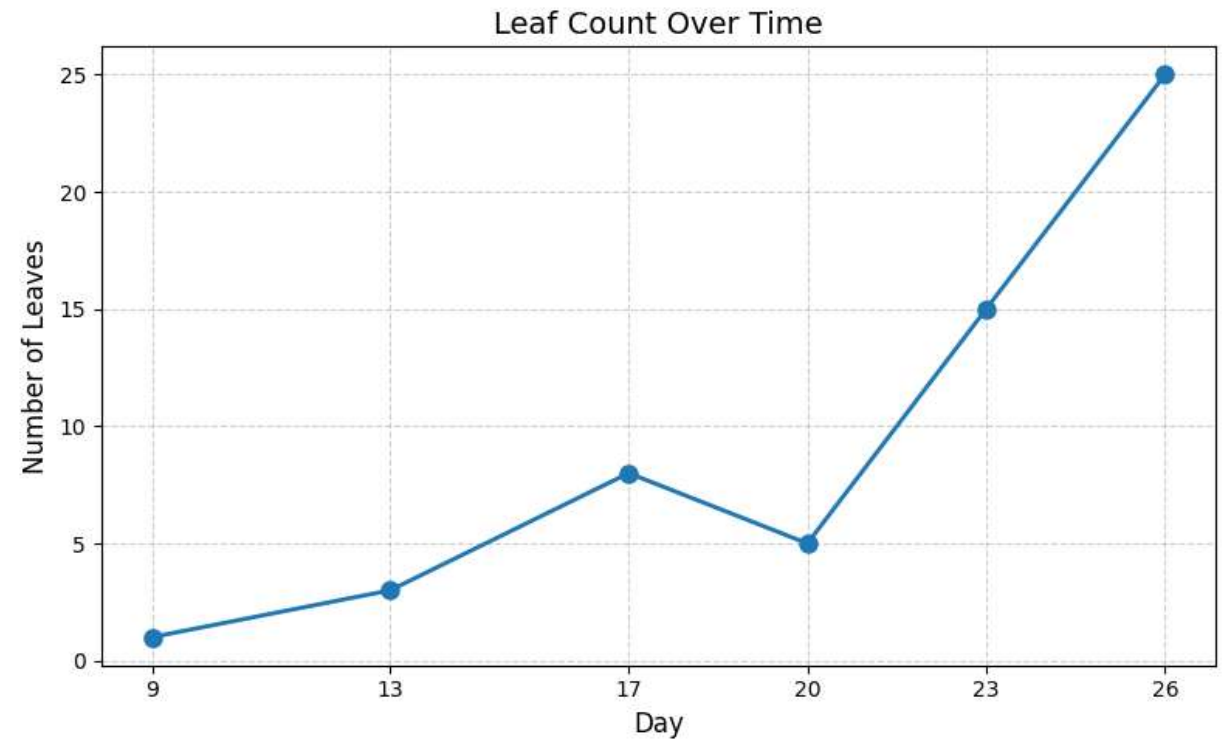




Results

- ✓ **Leaf count per day** – Number of leaves detected for each observation day
- ✓ **Individual leaf length** – Length of each leaf measured in pixels
- ✓ **Image segmentation** – Plant regions are separated accurately from the background
- ✓ **Skeletonization** – Leaves reduced to 1-pixel-wide structures for analysis
- ✓ **Bounding boxes** – Individual leaves identified and enclosed for visualization
- ✓ **Growth trend plot** – Leaf count plotted against time to show growth pattern.

Day001-09



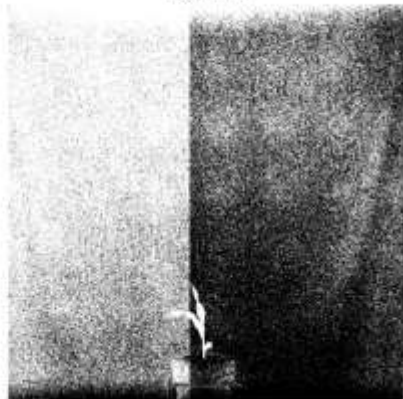
SUMMARY TABLE (sorted by day)

Day	Leaves	Lengths (px)
9	1	13
13	3	14, 12, 19
17	8	18, 19, 12, 38, 38, 17, 16, 35
20	5	18, 13, 19, 14, 12
23	15	21, 22, 13, 13, 14, 13, 21, 22, 27, 12, 18, 14, 18, 13, 28
26	25	19, 25, 25, 23, 16, 16, 14, 15, 14, 13, 17, 21, 18, 18, 15, 14, 18, 16, 27, 16, 13, 18, 13, 15, 15

Day_009 (7) — Original



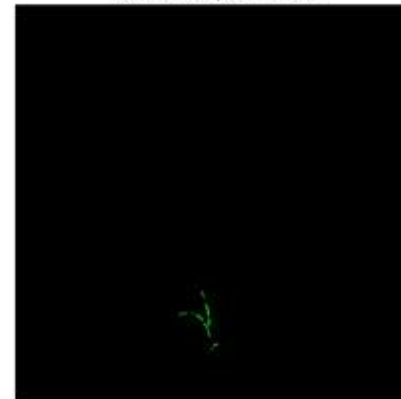
Lab Mask



Day_009 (7)
ExG Mask



Skeleton + Tips (1 leaves)



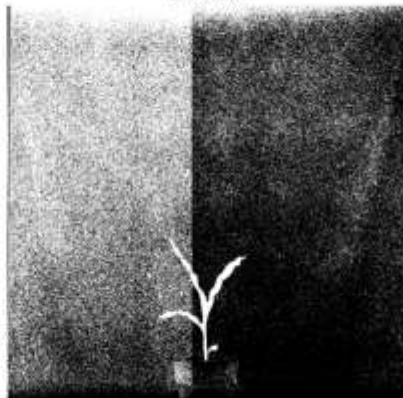
Leaf Bounding Boxes (1 detected)



Day_013 (7) — Original



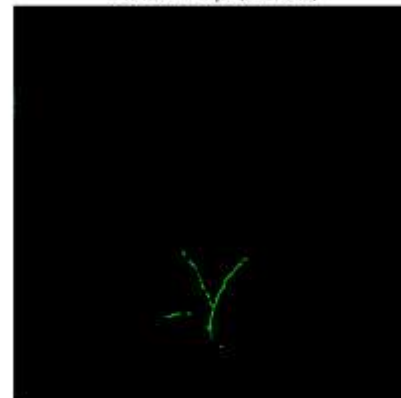
Lab Mask



Day_013 (7)
ExG Mask



Skeleton + Tips (3 leaves)



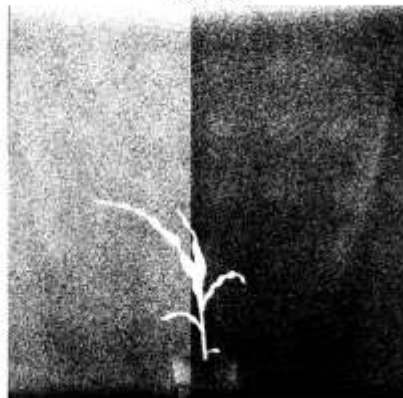
Leaf Bounding Boxes (3 detected)



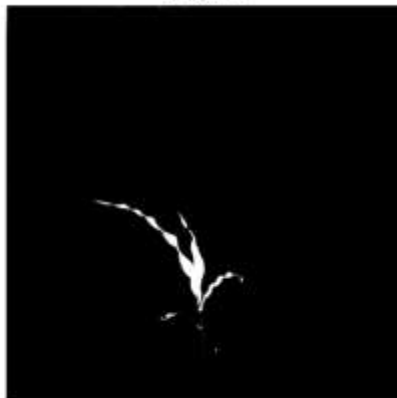
Day_017 (7) — Original



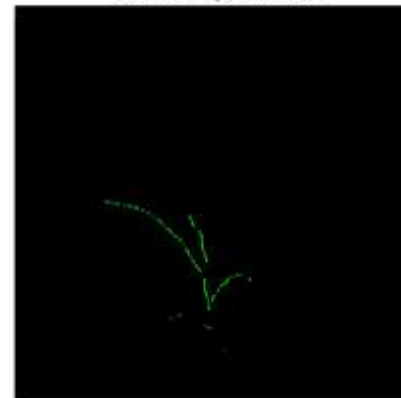
Lab Mask



Day_017 (7)
ExG Mask



Skeleton + Tips (8 leaves)



Leaf Bounding Boxes (8 detected)



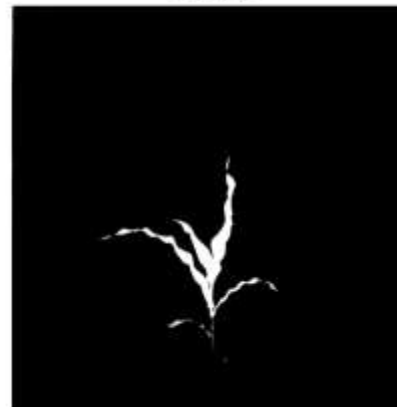
Day_020 (7) — Original



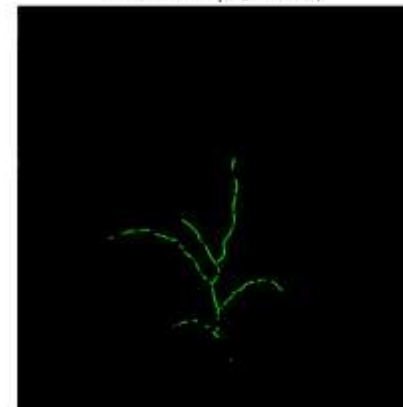
Lab Mask



Day_020 (7)
ExG Mask



Skeleton + Tips (5 leaves)



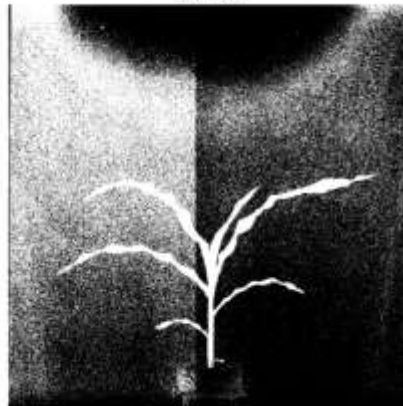
Leaf Bounding Boxes (5 detected)



Day_023 — Original



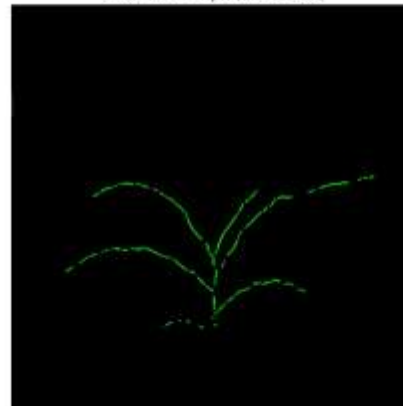
Lab Mask



Day_023
ExG Mask



Skeleton + Tips (15 leaves)



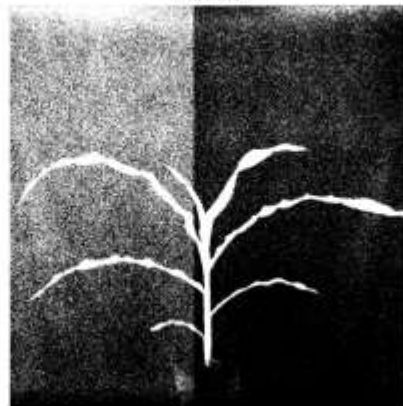
Leaf Bounding Boxes (15 detected)



Day_026 — Original



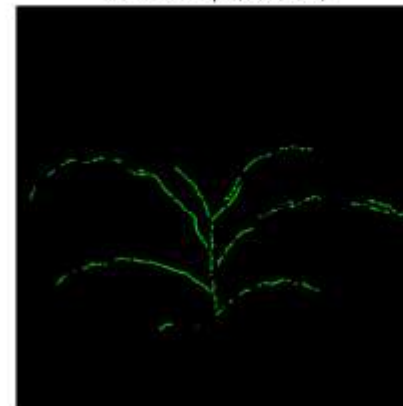
Lab Mask



Day_026
ExG Mask



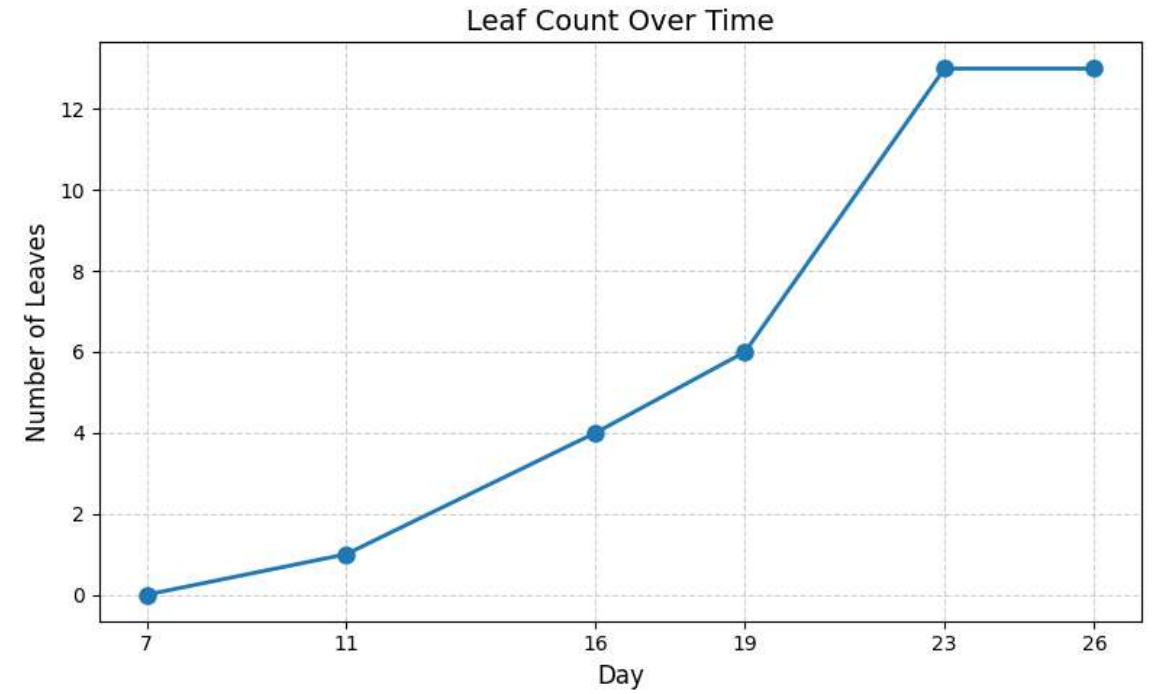
Skeleton + Tips (25 leaves)



Leaf Bounding Boxes (25 detected)



Day008-19



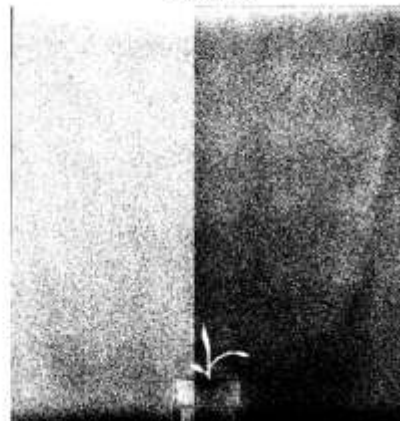
SUMMARY TABLE (sorted by day)

Day	Leaves	Lengths (px)
7	0	
11	1	12
16	4	14, 24, 13, 12
19	6	16, 15, 12, 17, 16, 12
23	13	33, 13, 13, 15, 19, 13, 22, 12, 19, 15, 17, 15, 17
26	13	21, 13, 18, 12, 17, 16, 12, 13, 13, 20, 13, 16, 19

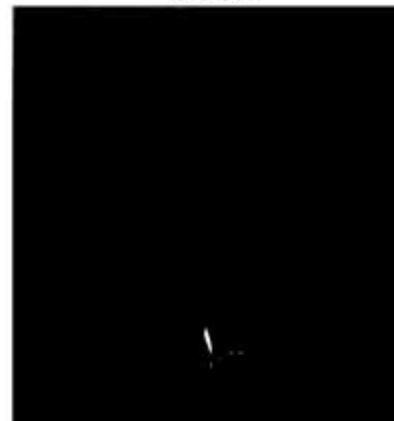
Day_007 — Original



Lab Mask



Day_007
ExG Mask



Skeleton + Tips (0 leaves)



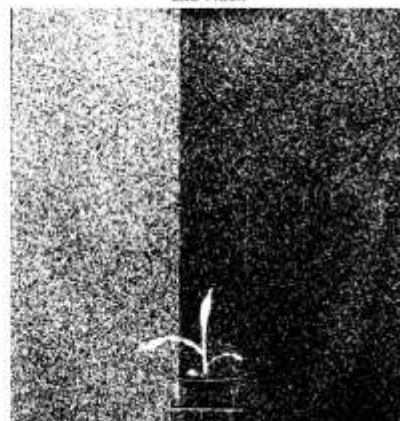
Leaf Bounding Boxes (0 detected)



Day_011 — Original



Lab Mask



Day_011
ExG Mask



Skeleton + Tips (1 leaves)



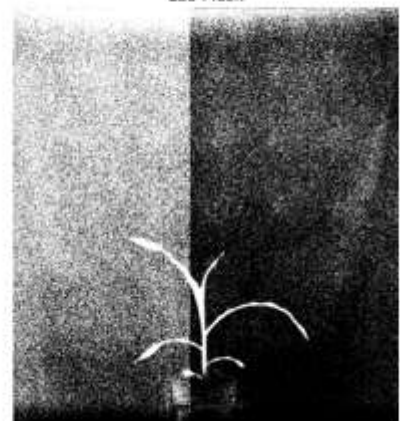
Leaf Bounding Boxes (1 detected)



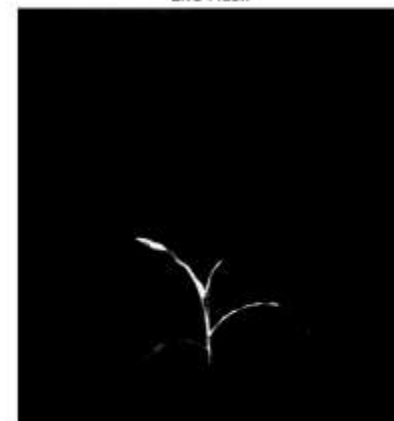
Day_016 — Original



Lab Mask



Day_016
ExG Mask



Skeleton + Tips (4 leaves)



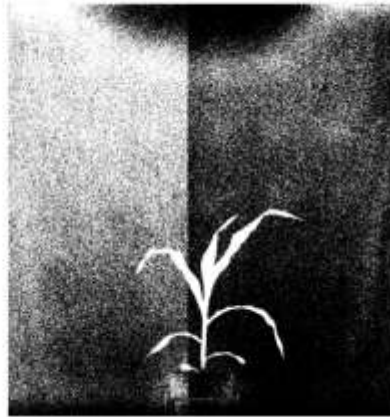
Leaf Bounding Boxes (4 detected)



Day_019 — Original



Lab Mask



Day_019
ExG Mask



Skeleton + Tips (6 leaves)



Leaf Bounding Boxes (6 detected)



Day_023 (1) — Original



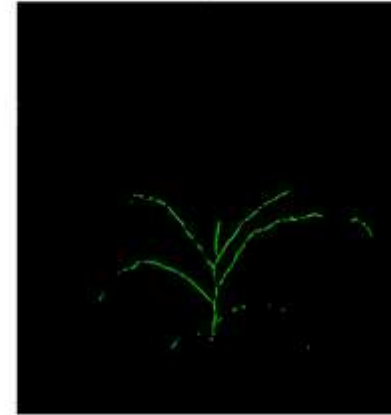
Lab Mask



Day_023 (1)
ExG Mask



Skeleton + Tips (13 leaves)



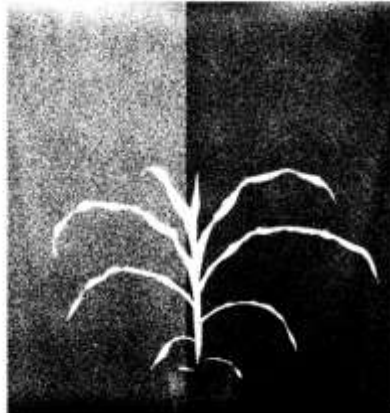
Leaf Bounding Boxes (13 detected)



Day_026 (1) — Original



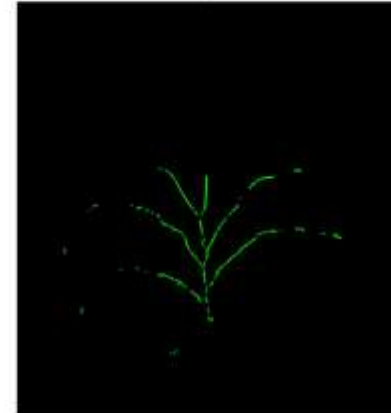
Lab Mask



Day_026 (1)
ExG Mask



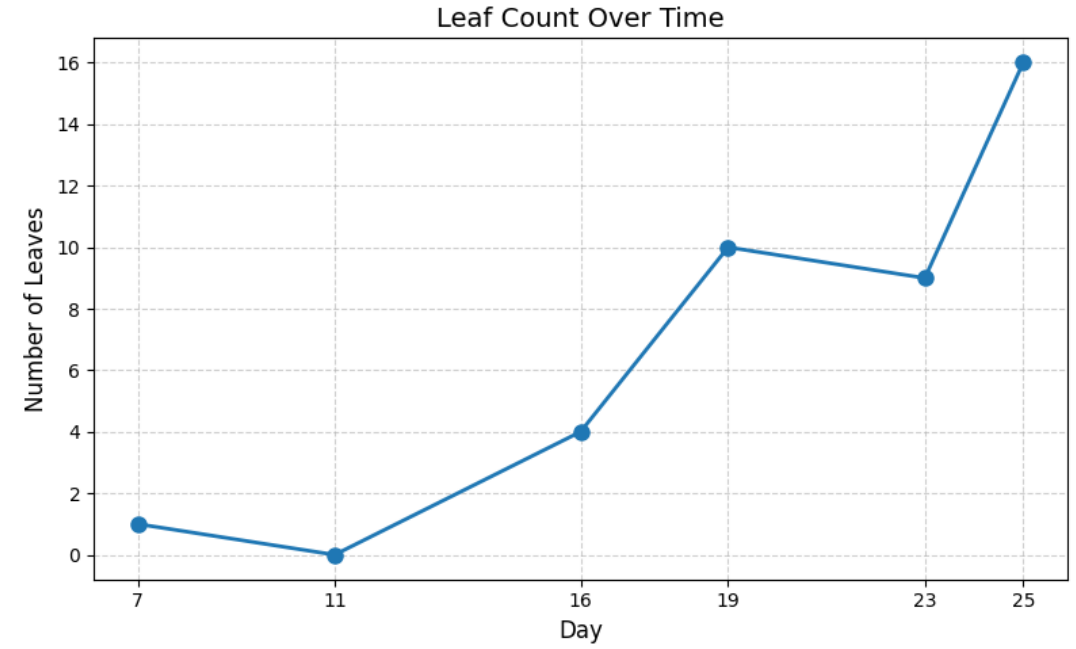
Skeleton + Tips (13 leaves)



Leaf Bounding Boxes (13 detected)



Day070-11



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SUMMARY TABLE (sorted by day)

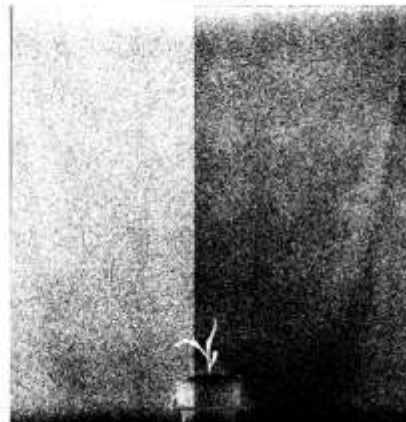
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Day	Leaves	Lengths (px)
7	1	13
11	0	
16	4	12, 16, 34, 19
19	10	12, 16, 14, 15, 12, 14, 14, 17, 26, 24
23	9	12, 13, 20, 41, 29, 13, 12, 13, 13
25	16	25, 25, 39, 20, 13, 31, 52, 14, 15, 20, 14, 17, 12, 13, 14, 12

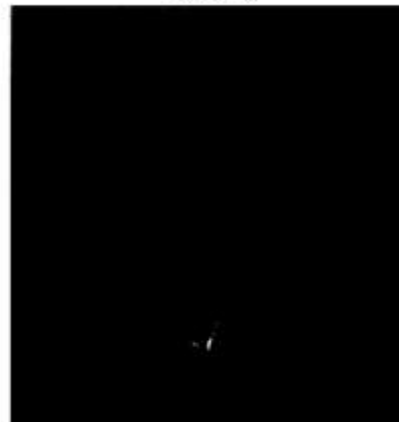
Day_007 (1) — Original



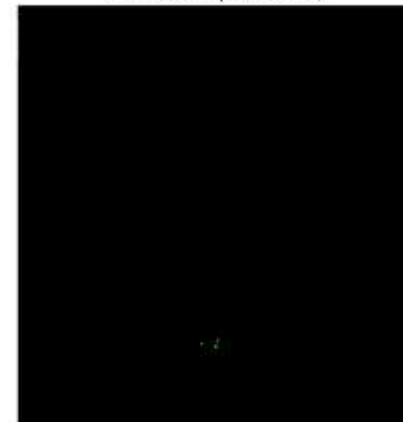
Lab Mask



Day_007 (1)
ExG Mask



Skeleton + Tips (1 leaves)



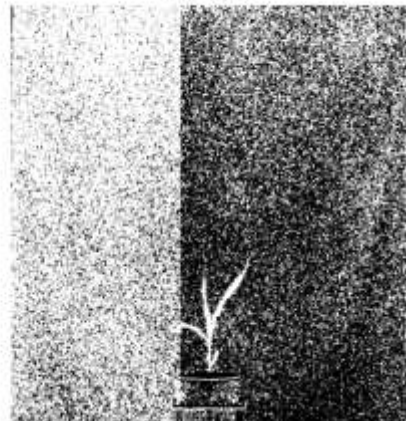
Leaf Bounding Boxes (1 detected)



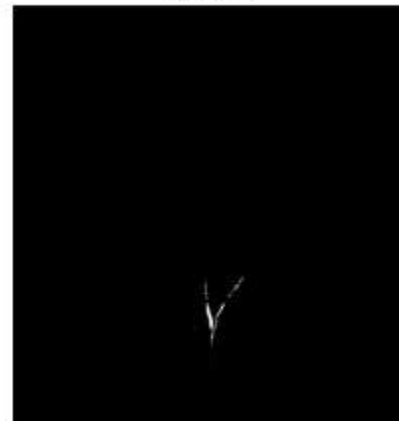
Day_011 (1) — Original



Lab Mask



Day_011 (1)
ExG Mask



Skeleton + Tips (0 leaves)



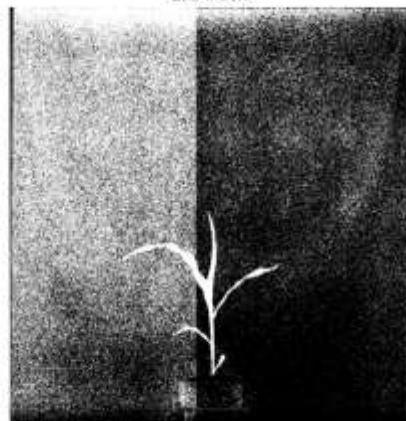
Leaf Bounding Boxes (0 detected)



Day_016 (1) — Original



Lab Mask



Day_016 (1)
ExG Mask



Skeleton + Tips (4 leaves)



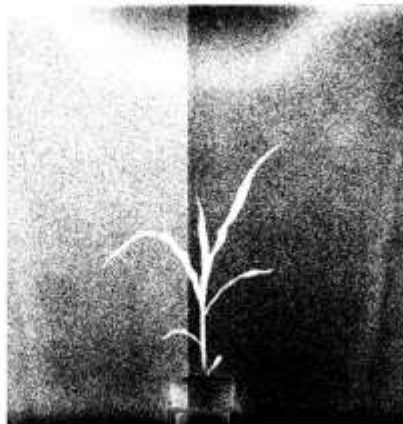
Leaf Bounding Boxes (4 detected)



Day_019 (1) — Original



Lab Mask



Day_019 (1)
ExG Mask



Skeleton + Tips (10 leaves)



Leaf Bounding Boxes (10 detected)



Day_023 (2) — Original



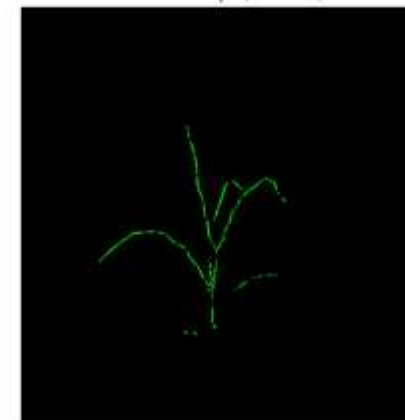
Lab Mask



Day_023 (2)
ExG Mask



Skeleton + Tips (9 leaves)



Leaf Bounding Boxes (9 detected)



Day_025 — Original



Lab Mask



Day_025
ExG Mask



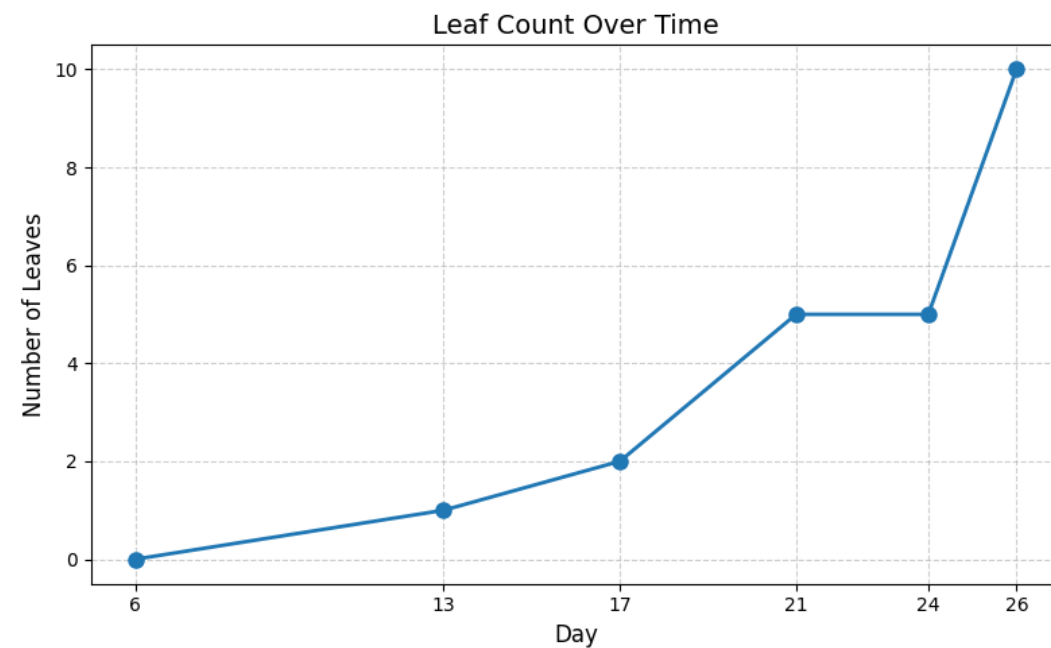
Skeleton + Tips (16 leaves)



Leaf Bounding Boxes (16 detected)



Day076-24



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SUMMARY TABLE (sorted by day)

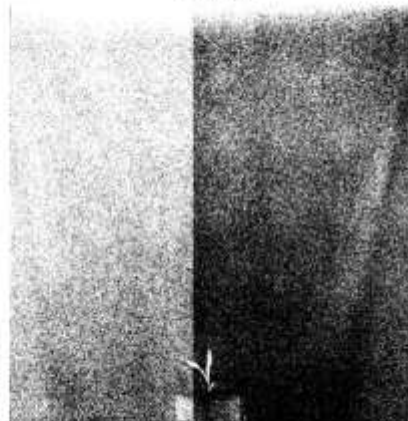
=====

Day	Leaves	Lengths (px)
6	0	
13	1	13
17	2	21, 16
21	5	19, 15, 16, 27, 16
24	5	17, 14, 22, 18, 13
26	10	18, 13, 12, 15, 17, 14, 22, 14, 31, 31

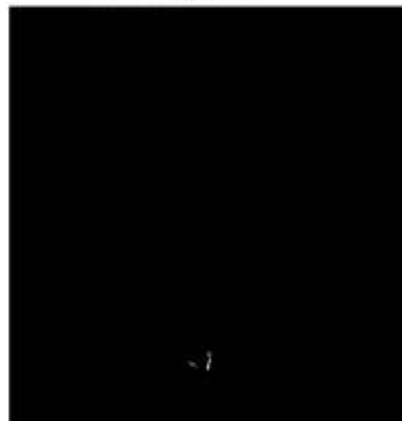
Day_006 — Original



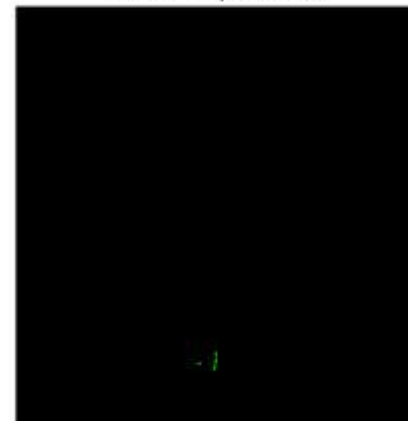
Lab Mask



Day_006
ExG Mask



Skeleton + Tips (0 leaves)



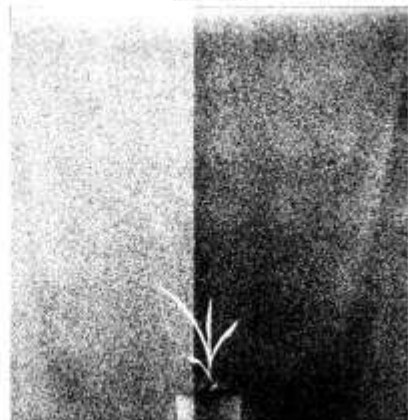
Leaf Bounding Boxes (0 detected)



Day_013 (8) — Original



Lab Mask



Day_013 (8)
ExG Mask



Skeleton + Tips (1 leaves)



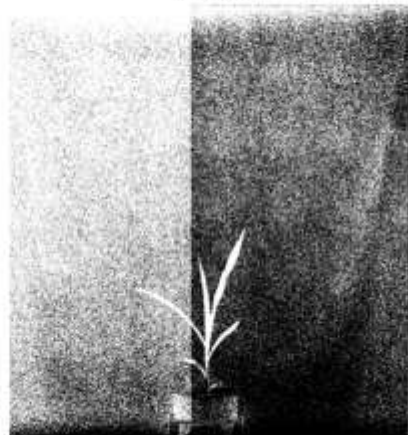
Leaf Bounding Boxes (1 detected)



Day_017 (8) — Original



Lab Mask



Day_017 (8)
ExG Mask



Skeleton + Tips (2 leaves)



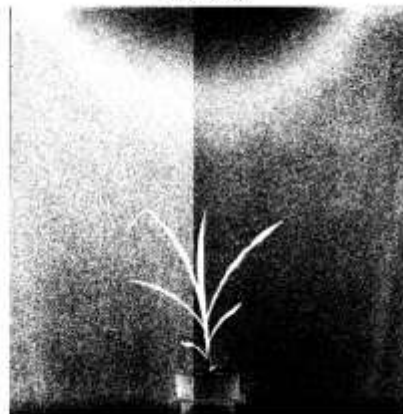
Leaf Bounding Boxes (2 detected)



Day_021 — Original



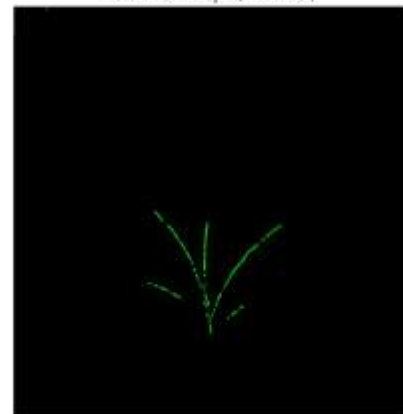
Lab Mask



Day_021
ExG Mask



Skeleton + Tips (5 leaves)



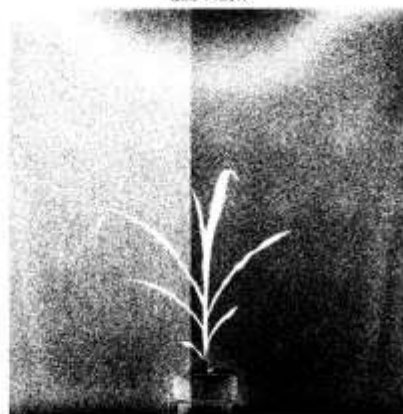
Leaf Bounding Boxes (5 detected)



Day_024 — Original



Lab Mask



Day_024
ExG Mask



Skeleton + Tips (5 leaves)



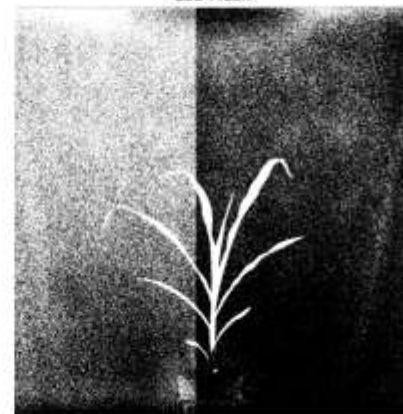
Leaf Bounding Boxes (5 detected)



Day_026 (2) — Original



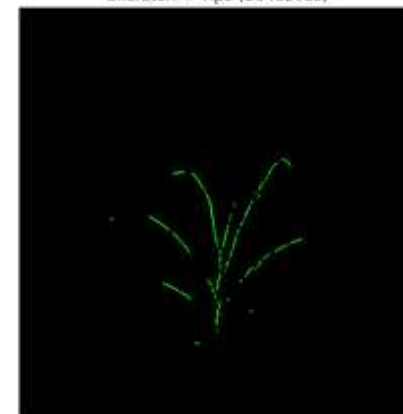
Lab Mask



Day_026 (2)
ExG Mask



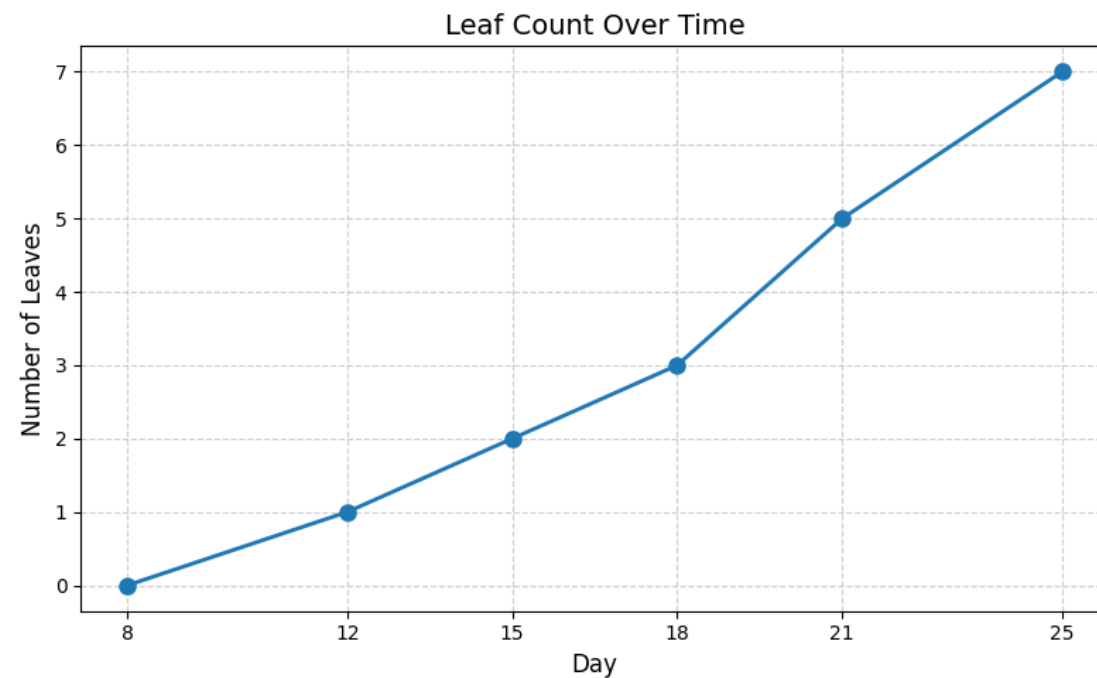
Skeleton + Tips (10 leaves)



Leaf Bounding Boxes (10 detected)



Day104-24



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SUMMARY TABLE (sorted by day)

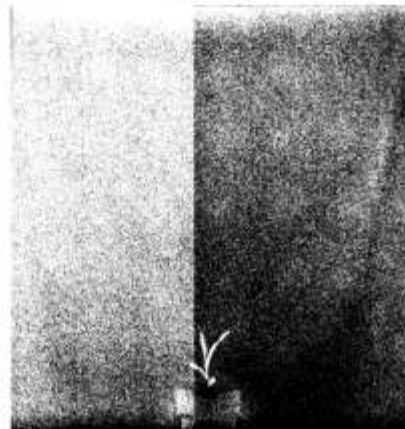
=====

Day	Leaves	Lengths (px)
8	0	
12	1	21
15	2	16, 17
18	3	14, 12, 17
21	5	13, 16, 15, 17, 20
25	7	12, 14, 15, 15, 15, 12, 13

Day_008 (1) — Original



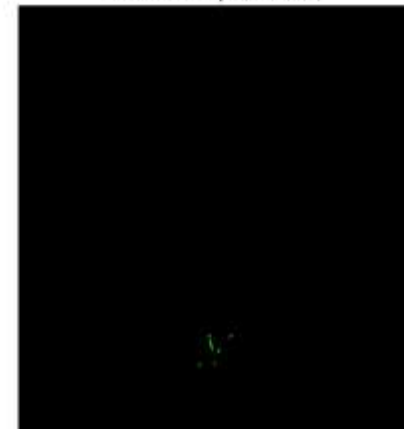
Lab Mask



Day_008 (1)
ExG Mask



Skeleton + Tips (0 leaves)



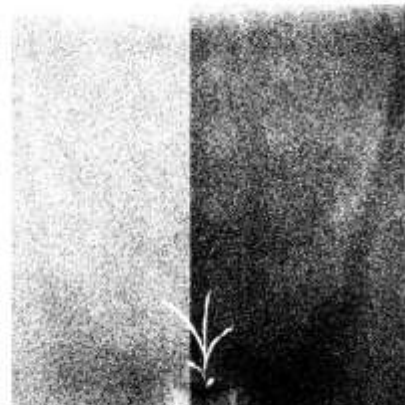
Leaf Bounding Boxes (0 detected)



Day_012 (1) — Original



Lab Mask



Day_012 (1)
ExG Mask



Skeleton + Tips (1 leaves)



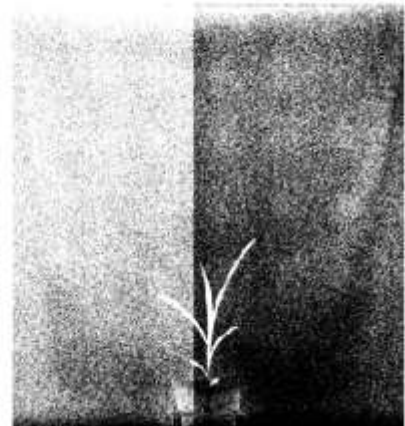
Leaf Bounding Boxes (1 detected)



Day_015 (1) — Original



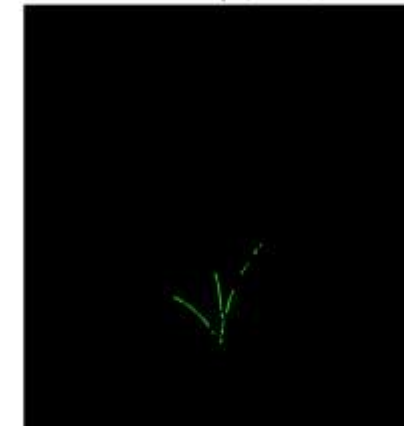
Lab Mask



Day_015 (1)
ExG Mask



Skeleton + Tips (2 leaves)



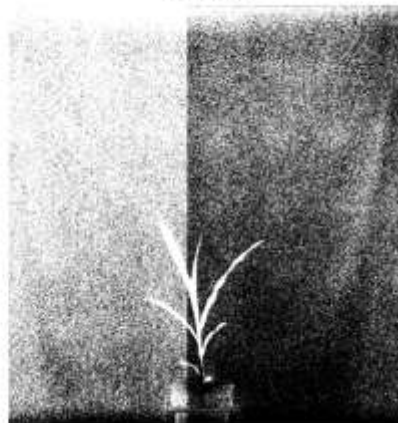
Leaf Bounding Boxes (2 detected)



Day_018 (1) — Original



Lab Mask



Day_018 (1)
ExG Mask



Skeleton + Tips (3 leaves)



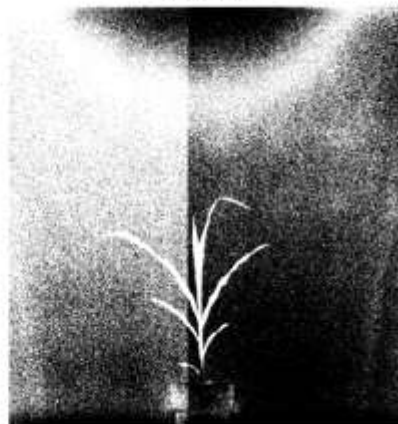
Leaf Bounding Boxes (3 detected)



Day_021 (2) — Original



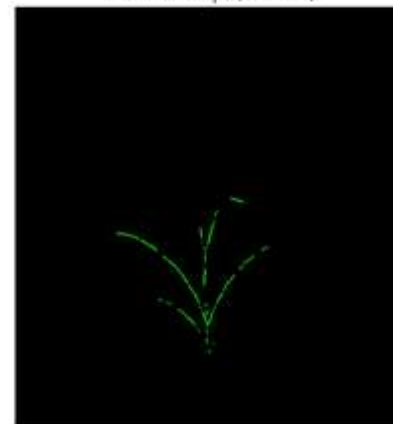
Lab Mask



Day_021 (2)
ExG Mask



Skeleton + Tips (5 leaves)



Leaf Bounding Boxes (5 detected)



Day_025 (2) — Original



Lab Mask



Day_025 (2)
ExG Mask



Skeleton + Tips (7 leaves)



Leaf Bounding Boxes (7 detected)



Future Applications

- Enables development of **high-yield, stress-resistant crops**
- Supports **precision agriculture** through real-time plant monitoring
- Helps in **early disease detection** using image analysis
- Use in **greenhouse automation and smart farming systems**
- Assists in **crop growth prediction and yield estimation.**

